# Eyad Eldokany

**AI Engineer**

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## Summary

AI Engineer specializing in LLMs, Generative AI, Computer Vision, and Deep Learning. Experienced in designing and deploying production-oriented AI systems using FastAPI, AWS Bedrock, RAG, agentic workflows, Docker, and Kubernetes. Built multimodal healthcare and AI developer platforms combining intelligent models with scalable backend and cloud infrastructure.

## Education

**Egyptian E-Learning University (EELU)** — Alexandria, Egypt
*B.Sc. Computer Science and Artificial Intelligence · CGPA: 3.74/4.0 (Excellent)* — *Sep 2022 – Jun 2026*

## Experience

**AI Engineer | Independent Projects** — Remote
*Self-Directed Projects & Open Source* — *2025 – Present*

- Designed and deployed AI solutions across LLM, computer vision, and machine learning use cases, using FastAPI, Docker, AWS, and modern AI frameworks.
- Built production-oriented AI APIs, RAG pipelines, agentic workflows, and computer vision systems tailored to client requirements.
- Developed end-to-end solutions spanning model development, backend APIs, deployment, and monitoring.

**National Telecommunication Institute (NTI)** — Egypt
*Computer Vision Trainee* — *Aug 2025 – Sep 2025*

- Engineered real-time object detection and classification pipelines using YOLOv11 and OpenCV, optimizing inference speed through confidence threshold tuning and NMS adjustments.
- Built and trained custom YOLO-based models; handled data preprocessing, annotation, and augmentation.
- Implemented OpenCV preprocessing utilities for edge detection, morphological operations, and feature extraction across multiple detection tasks.

**DEPI (Digital Egyptian Pioneers Initiative)** — Egypt
*AI & Data Science Trainee* — *Oct 2024 – May 2025*

- Developed predictive ML models to enhance data-driven decision-making.
- Performed EDA and feature engineering pipelines in Python, Pandas, and scikit-learn, delivering interpretable results for non-technical stakeholders.
- Earned certified Data Scientist qualification through the national DEPI program.

**National Telecommunication Institute (NTI)** — Egypt
*AI & ML Trainee* — *Aug 2024 – Sep 2024*

- Built ML models for trend prediction and anomaly detection on structured datasets.
- Gained hands-on experience with model training, evaluation, and hyperparameter tuning using Scikit-Learn and TensorFlow.

## Projects

**HealthBridge** | *AWS EKS, FastAPI, pgvector, Bedrock, MLOps* — Graduation Project

- Built a cloud-native medical AI platform on AWS EKS with 4 FastAPI microservices, a custom ACE agentic pipeline (Reflector, Curator, Deduplication agents), and a 27-source RAG system using pgvector and Deep Research Mode for accurate clinical knowledge retrieval.
- Integrated a dual-model computer vision layer with AWS Bedrock LLM reasoning to enable end-to-end multimodal patient assessment across consultation, diagnosis, and automated medical summary workflows.
- Deployed full MLOps observability stack (Prometheus, Loki, Grafana) and async RabbitMQ orchestration across all services — delivering a production-ready, bilingual Arabic/English healthcare AI system.
- **Grade:** A+ (Excellent).

**CodeNova — AI Coding Assistant** | *Python, FastAPI, Supabase, Docker, Fly.io* — Project Live URL ↗

- Built and deployed a full-stack AI coding assistant with a real-time streaming chat interface, agentic tool-calling loop, and per-conversation isolated project workspaces. Features 15 agent tools (file I/O, Python execution, testing, linting, doc search), Supabase Auth/PostgreSQL backend, SSE streaming, and a responsive single-file frontend with markdown rendering. Deployed via Docker on Fly.io with GitHub Actions CI/CD.

**Multimodal Medical AI Platform** | *Python, Flask, Docker, Scikit-Learn, XGBoost* Project Link ↗

- Built and deployed a modular healthcare platform integrating 8 ML diagnosis models (Heart Disease, Diabetes, Liver Disorder, Autism, Anemia, and COVID-19 Logistic & XGBoost variants). Packaged models using Flask APIs and Docker containers for fast, portable deployment.

**Real-Time Edge AI Attendance** | *Python, YOLOv8, ByteTrack, InsightFace, FAISS, SQLite* Project Link ↗

- Real-time Pipeline: Engineered a 24/7 RTSP video system using YOLOv8 & ByteTrack for multi-person tracking and line-crossing detection with auto-reconnection.
- Face Recognition & Vector Search: Integrated InsightFace with FAISS L2 indexing for real-time identification, confidence scoring, and fallback verification.
- Active Enrollment & Thread-Safe Logging: Built an automated active learning pipeline for unknown faces and a thread-safe SQLite (WAL) engine with background Excel sync.

**Face Recognition Sign-In System** | *Python, Flask, OpenCV, Railway, Qdrant* Project Link ↗

- Developed a real-time facial recognition authentication system with Qdrant-based identity retrieval and database integration.

**Alzheimer MRI Detection** | *PyTorch, ResNet-18, Flask, Torchvision* Project Link ↗

- Trained a ResNet-18 classifier on the Alzheimer's MRI dataset (4 classes) achieving 98% validation accuracy via transfer learning. Built a lightweight Flask web interface for real-time diagnosis.

**Traffic Sign Detection** | *Python, YOLOv11, PyTorch, OpenCV, FastAPI* Project Link ↗

- Developed an intelligent real-time traffic sign detection system optimized for embedded and edge deployment.

**YOLOv11m Industrial Clamp Detection** | *Python, YOLOv11m, Roboflow, ONNX* Project Link ↗

- Developed an industrial clamp detection system using YOLOv11m, building and annotating a custom dataset entirely on Roboflow. Achieved strong detection accuracy (mAP@0.5: 75.48%) with deployment support via ONNX/TensorRT for real-time performance on NVIDIA GPUs.

## Technical Skills

**AI / ML**: PyTorch, TensorFlow, Keras, scikit-learn, YOLOv11, OpenCV, HuggingFace, MLflow
**LLMs & Agents**: AWS Bedrock, LangChain, LlamaIndex, RAG, ACE Framework, Prompt Engineering, Qdrant, Chroma DB, FAISS, Milvus, Cohere, NVIDIA NIM, OpenAI
**Languages**: Python, Java, C++, C, JavaScript, HTML/CSS, SQL
**Backend & APIs**: FastAPI, Flask, Streamlit, Gradio, REST APIs, RabbitMQ, Celery
**Cloud & DevOps**: AWS, Docker, Vercel, Git, DynamoDB, Railway, MLOps, Kubernetes/EKS
**Data Science**: NumPy, Pandas, Matplotlib, Plotly, SciPy, XGBoost

## Certifications

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|---|---|
| **Advanced RAG with Vector Databases and Retrievers** \| IBM | 2026 |
| **Vector Databases for RAG: An Introduction** \| IBM | 2026 |
| **Build RAG Applications: Get Started** \| IBM | 2026 |
| **Develop Generative AI Applications: Get Started** \| IBM | 2026 |
| **Deep Learning with Keras and TensorFlow** \| IBM | 2025 |
| **Introduction to Deep Learning & Neural Networks with Keras** \| IBM | 2025 |
| **Getting Started with Deep Learning** \| NVIDIA | 2025 |
| **Machine Learning with Python** \| IBM | 2025 |
| **CCNA Routing and Switching** \| Cisco | 2025 |
| **Data Science and Machine Learning** \| Microsoft Student Club, EELU | 2024 |
| **HCIA-AI V3.5** \| Huawei | 2024 |
| **Security, Compliance, and Identity Fundamentals** \| Microsoft | 2024 |
| **CCNAv7** \| Cisco | 2024 |

## Languages

**Arabic**: Native · **English**: Fluent · **German**: Basic